计算方法B
12.18应交上机作业

宋思逸
2400010816

2025 年 12 月 15 日

1 问题

最速下降，Newton，拟Newton(SR1,DFP,BFGS)，BB方法的实现
对最优化问题

$$min \sum_{i=1}^{m} r_i^2(x) \tag{1}$$

选择不同规模，即不同n或m，利用并好的程序进行计算，其中$r_i(x)$如下

- Watson函数

$$r_i(x) = \sum_{j=2}^{n}(j-1)x_j t_i^{j-2} - (\sum_{j=1}^{n} x_j t_i^{j-1})^2 - 1 \tag{2}$$

其中$t_i = i/29, 1 \leq i \leq 29, r_{30}(x) = x_1, r_{31} = x_2 - x_1^2 - 1, 2 \leq n \leq 31$初始点可选为$x^{(0)} = (0, \cdots, 0)^T$

- Discrete boundary value函数

$$r_i(x) = 2x_i - x_{i-1} - x_{i+1} + \frac{h^2(x_i + t_i + 1)^3}{2} \tag{3}$$

其中$h = \frac{1}{n+1}, t_i = ih, x_0 = x_{n+1} = 0, m = n$初始点可以选为

$$x^{(0)} = (t_1(t_1 - 1), \cdots, t_n(t_n - 1))^T$$

通过计算，可以进行关于线搜索不同搜索准则，不同插值方法之间的比较，也可以通过输出的信息，进行不同Newton型方法有效性的比较。输出信息可以包含算法的迭代次数、函数调用次数、导数调用次数及CPU时间。

2 代码实现

2.1 最速下降法

- 给出$x_0 \in \mathbb{R}, \varepsilon > 0, k := 0$
- 若终止条件满足，则迭代停止。
- 计算$d_k = -g_k$
- 一维精确线搜索求α_k
- $x_{k+1} := x_k + \alpha_k + d_k, k := k + 1$

2.2 基本Newton方法

- 给出$x_0 \in \mathbb{R}, \varepsilon > 0, k := 0$
- 若终止准则满足，则输出有关信息，停止迭代
- 由$G_k d = -g_k$计算$d_k = -G_k^{-1} g_k$
- $x_{k+1} := x_k + d_k, k := k + 1$

不过代码中给出的是Gauss-Newton方法，这一方法在书中并没有介绍

2.3 拟Newton方法

- 给定$x_0 \in \mathbb{R}$，对称正定阵$H_0 \in \mathbb{R}^{n \times n}, \varepsilon > 0, k := 0$
- 若终止准则满足，则输出有关信息，停止迭代
- 计算$d_k = -H_k g_k$，其中$B_{k+1} s_k = y_k, H_{k+1} y_k = s_k, s_k = x_{k+1} - x_k, y_k = g_{k+1} - g_k$且有$G_{k+1} s_k = y_k + O(||s_k||^2)$

2.3.1 SR1

$$H_{k+1}^{SR1} = H_k + \frac{(s_k - H_k y_k)(s_k - H_k y_k)^T}{(s_k - H_k y_k)^T y_k} \tag{4}$$

2.3.2 DFP

$$H_{k+1}^{DFP} = H_k + \frac{s_k s_k^T}{s_k^T y_k} - \frac{H_k y_k y_k^T H_k}{y_k^T H_k y_k} \tag{5}$$

2.3.3 BFGS

$$H_{k+1}^{BFGS} = H_k + (1 + \frac{y_k^T H_k y_k}{y_k^T s_k})\frac{s_k s_k^T}{y_k^T s_k} - (\frac{s_k y_k^T H_k + H_k y_k s_k^T}{y_k^T s_k}) \tag{6}$$

2.4 BB

考虑负梯度迭代

$$x_{k+1} = x_k - \alpha_k g_k$$

其中$g_k = Gx_k + b$，下面考虑如何选取α_k

2.4.1 BB1

$$\alpha_k^{BB1} = \frac{g_{k-1}^T g_{k-1}}{g_{k-1}^T G g_{k-1}} \tag{7}$$

2.4.2 BB2

$$\alpha_k^{BB2} = \frac{g_{k-1}^T G g_{k-1}}{g_{k-1}^T G^2 g_{k-1}} \tag{8}$$

3 结果输出

```
>> optimization_methods_demo
=== demo: Watson n=6 ===
Method: steepest
Method: Steepest Descent
iter = 200, f = 2.572187e+01, ||g|| = 1.130e+02, time = 0.0719 s
calls: f=0, grad=0, jac=0, residual=0

Method: gn
```

```
Method: Gauss-Newton
iter = 200, f = 7.636590e+00, ||g|| = 1.595e+01, time = 0.0949 s
calls: f=0, grad=0, jac=0, residual=0

Method: bfgs
Method: Quasi-Newton BFGS
iter = 200, f = 1.969329e+01, ||g|| = 5.024e+01, time = 0.0784 s
calls: f=0, grad=0, jac=0, residual=0

Method: dfp
Method: Quasi-Newton DFP
iter = 200, f = 1.811462e+01, ||g|| = 5.510e+01, time = 0.0760 s
calls: f=0, grad=0, jac=0, residual=0

Method: sr1
Method: Quasi-Newton SR1
iter = 200, f = 1.979140e+01, ||g|| = 8.116e+01, time = 0.0716 s
calls: f=0, grad=0, jac=0, residual=0

Method: bb1
Method: Barzilai-Borwein type 1
iter = 200, f = 2.438445e+03, ||g|| = 1.740e+03, time = 0.0169 s
calls: f=0, grad=0, jac=0, residual=0

Method: bb2
Method: Barzilai-Borwein type 2
iter = 200, f = 7.038210e+01, ||g|| = 1.452e+00, time = 0.0153 s
calls: f=0, grad=0, jac=0, residual=0


=== demo: Discrete BVP n=50 ===
Method: Gauss-Newton
iter = 3, f = 9.001195e-18, ||g|| = 5.936e-11, time = 0.0343 s
calls: f=0, grad=0, jac=0, residual=0
```

```
Method: Quasi-Newton BFGS
iter = 97, f = 8.117166e-19, ||g|| = 3.412e-10, time = 0.0182 s
calls: f=0, grad=0, jac=0, residual=0
```